

Fausto Giunchiglia, Leonardo Javier Malcotti

Computational Logic Handouts

October 27, 2025

Contents

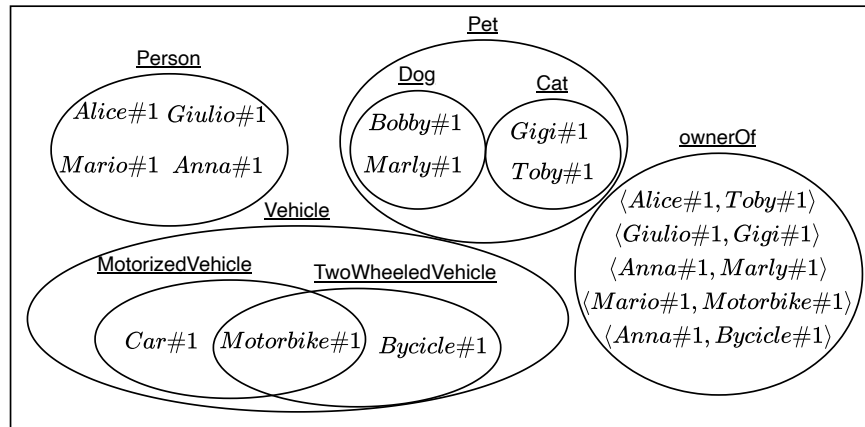
1	LoD	5
1.1	LoD models, theories and interpretation function	5
1.2	Translation across formalization levels	7
1.3	Solutions	9

Chapter 1

LoD

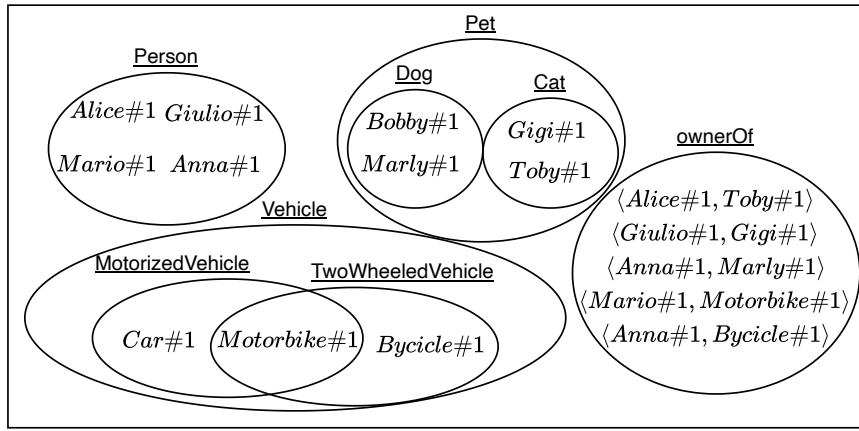
1.1 LoD models, theories and interpretation function

Exercise 1.1 Given the following intended model, assuming that the interpretation function is built by doing syntactic matching between the tokens used in the language and the names given to elements and sets of the model, determine the set of entities represented by the assertions in the language of etype percepts.



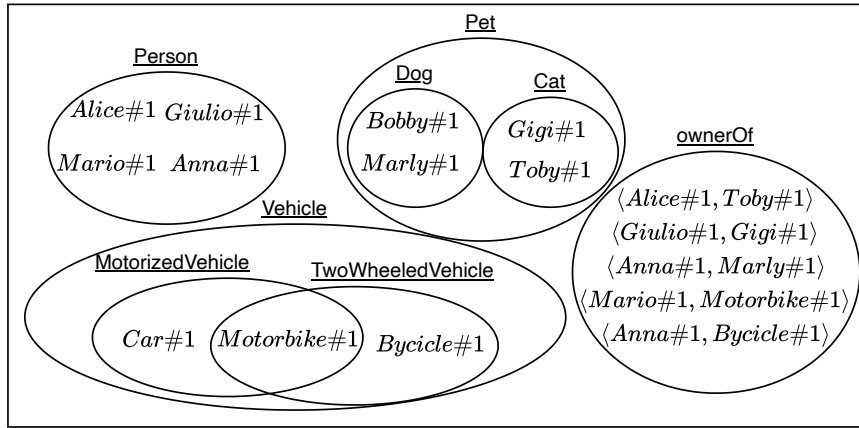
1. *MotorizedVehicle*
2. *Pet*
3. $\exists \text{ownerOf.Pet}$
4. $\forall \text{ownerOf.Cat}$
5. $\exists \text{ownerOf}.\top$
6. $\forall \text{ownerOf}.\perp$
7. $\exists \text{ownerOf}.\perp$

Exercise 1.2 Given the following intended model, assuming that the interpretation function is built by doing syntactic matching between the tokens used in the language and the names given to elements and sets of the model, determine the set of entities represented by the assertions in the language of composite etype percepts.



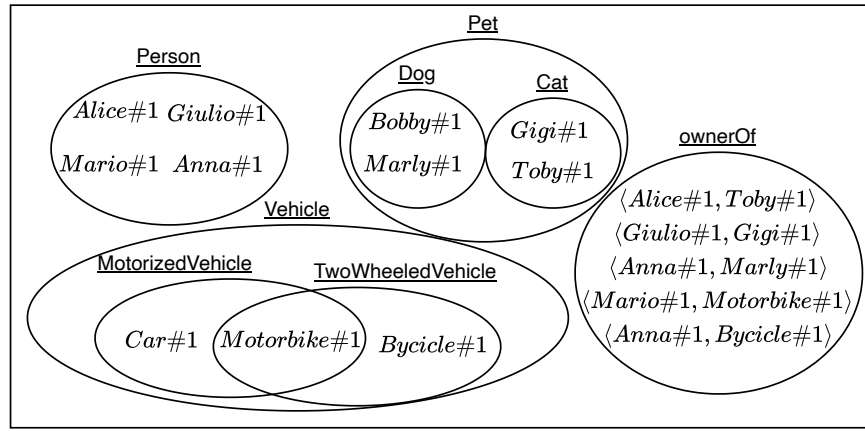
1. $MotorizedVehicle \sqsubset (Person \sqcap \forall ownerOf.Vehicle)$
2. $Person \sqcap \exists ownerOf.(MotorizedVehicle \sqcap TwoWheeledVehicle)$
3. $\neg Vehicle \sqcap \neg Dog \sqcap \forall ownerOf.\perp$
4. $\exists ownerOf.(Pet \sqcap \neg Cat)$
5. $Person \sqcap \neg \forall ownerOf.Cat$
6. $Person \sqcap \exists ownerOf.\neg Pet$
7. $Person \sqcap \exists ownerOf.(Vehicle \sqcap \neg TwoWheeledVehicle)$

Exercise 1.3 Given the following intended model, assuming that the interpretation function is built by doing syntactic matching between the tokens used in the language and the names given to elements and sets of the model, determine if the following statements in LoD reflect correctly the intended model.



1. $MotorizedVehicle \sqsubseteq Vehicle$
2. $\exists ownerOf.Pet \sqsubseteq Person$
3. $\exists ownerOf.Pet \equiv Person$
4. $Dog \sqcup Cat \equiv Pet$
5. $\forall ownerOf.Vehicle \sqsubseteq Person$
6. $\neg \forall ownerOf.Vehicle \sqsubseteq Person$
7. $Person \sqcap Pet \sqsubseteq \top$

Exercise 1.4 Given the following intended model, assuming that the interpretation function is built by doing syntactic matching between the tokens used in the language and the names given to elements and sets of the model, determine which of the following statements on LoD theories are true.



- $Pet \equiv Dog \sqcup Cat$
1. $MotorizedVehicle \sqsubseteq Vehicle$
 $TwoWheeledVehicle \sqsubseteq Vehicle$
 $Person \sqsubseteq \exists ownerOf.\top$ Is a valid theory with respect to proposed model.
 2. $Dog \sqsubseteq Pet$
 $Cat \sqsubseteq Pet$
 $Dog \perp Cat$ Is a valid theory with respect to proposed model.
 $MotorizedVehicle \sqsubseteq Vehicle$
 $TwoWheeledVehicle \sqsubseteq Vehicle$
 $Person \sqsubseteq \exists ownerOf.(Pet \sqcup Vehicle)$
 3. $Pet \sqsubseteq \neg Person$ Is a valid theory with respect to proposed model.
 $MotorizedVehicle \sqsubseteq Vehicle$
 $TwoWheeledVehicle \sqsubseteq Vehicle$
 $Person \sqsubseteq \exists ownerOf.\top$
 4. $Pet \equiv Dog \sqcup Cat$ Is a valid theory with respect to proposed model.
 $MotorizedVehicle \sqsubseteq Vehicle$
 $TwoWheeledVehicle \sqsubseteq Vehicle$
 $Person \sqsubseteq \forall ownerOf.Pet$
5. In a LoD theory we can represent that *Alice#1* is a *Person*.

1.2 Translation across formalization levels

Exercise 1.5 Indicate which of the formalizations in the various sub-languages of the Logic of Descriptions (LoD) of the natural language sentences below is correct (one or more):

1. The formalization of “The entities that live in a building” in the “Language of composite Etype percepts” is $\exists livesIn.Building$.
2. The formalization of “The entities that only talk to those entities that have no friends” in the “Language of etype percepts” is $\forall talksTo.(\exists hasFriend.\perp)$.
3. The formalization of “A cook is a person who works in a restaurant.” in the “Language of Composite Etype Percepts” is $Cook \sqsubseteq Person \sqcap \exists worksIn.Ristoraaurant$.
4. The formalization of “Vegetarian dishes are precisely those dishes that do not contain pork.” in the “Language of Definitions” is $VegetarianDishes \equiv Dishes \sqcap \neg \forall contain.Pork$
5. The formalization of “Smartphones are personal devices used for calling and messaging.” in the “Language of Definitions” is $Smartphone \sqsubseteq Devices \sqcap Personal \sqcap \exists usedFor.(Calling \sqcap Messaging)$

6. The formalization of “Smartphones are personal devices used for calling and messaging.” in the “Language of Definitions” is $Smartphone \sqsubseteq Devices \sqcap Personal \sqcap \exists usedFor.(Calling \sqcup Messaging)$
7. The formalization of “Vehicles that have a battery that cannot be recharged by pedaling” in the “Language of Composite Etype Percepts” is $Vehicles \sqcap \exists hasBattery.(Battery \sqcap \neg \exists rechargedBy.Pedaling)$

Exercise 1.6 Indicate which of the formalizations in the different sublanguages used by Description Logic (DL) of the following sentences in natural language are correct (one or more):

1. The formalization of “Entities that write only in cursive” in “Language of etype percepts” is: $\exists writesIn.Cursive$
2. The formalization of “A whiteboard is a type of white board on which one writes with markers” in “Language of composite etype percepts” is $Whiteboard \sqsubseteq Board \sqcap \exists color.White \sqcap \exists writesOnWith.Marker$
3. The formalization of “A Buddhist monk is a person who lives only with people who practice Buddhism” in “Language of Definitions” is $BuddhistMonk \sqsubseteq Person \sqcap \forall livesWith.(Person \sqcap \exists practices.Buddhism)$
4. The formalization of “The set of all textbooks with hard and soft covers” in “Language of composite etype percepts” is $Textbook \sqcap \exists hasCover.(Hardcover \sqcup Softcover)$
5. The formalization of “The set of all textbooks with hard and soft covers” in “Language of composite etype percepts” is $Textbook \sqcap \exists hasCover.(Hardcover \sqcap Softcover)$

Exercise 1.7 Indicate which of the formalizations in the different sublanguages used by the Logic of Descriptions (LOD) of the following sentences in natural language are correct (one or more):

1. The formalization of “Entities that only ride bicycles” in the “Language of composite Etype percepts” is $\forall ride.Bicycle$
2. The formalization of “entities that know who does not drink alcohol” in the “Language of etype percepts” is $\exists knows.(\forall drinks.\neg Alcohol)$
3. The formalization of “A researcher is a person with a degree” in the “Language of Composite Etype Percepts” is $Researcher \sqsubseteq Person \sqcap \exists has.Degree$
4. The formalization of “Oviparous animals are precisely those animals that lay only eggs” in the “Language of Definitions” is $Oviparous \equiv Animal \sqcap \exists lays.Eggs$
5. The formalization of “Pencils are used for writing and drawing” in the “Language of Definitions” is $Pencil \sqsubseteq \exists usedFor.(Writing \sqcap Drawing)$
6. The formalization of “Pencils are used for writing and drawing” in the “Language of Definitions” is $Pencil \sqsubseteq \exists usedFor.(Writing \sqcup Drawing)$
7. The formalization of “Watches whose springs are wound by the movement of the wrist” in the “Language of Composite Etype Percepts” is $Watch \sqcap \exists contains.(Spring \sqcap \exists woundBy.WristMovement)$

Exercise 1.8 Indicate which of the formalizations in the different sublanguages used by the Logic of Descriptions (LOD) of the following sentences in natural language are correct (one or more):

1. The formalization of “Entities that speak Spanish” in “Language of etype percepts” is: $\forall speaksIn.Spanish$
2. The formalization of “A jug is a container used to hold water” in “Language of composite etype percepts” is $Jug \sqsubseteq Container \sqcap \exists contains.Water$
3. The formalization of “A fan is an electrical appliance used in hot places” in “Language of Definitions” is $Fan \sqsubseteq Appliance \sqcap \forall usedIn.(Place \sqcap \exists hasTemperature.Hot)$
4. The formalization of “The set of all mechanical and membrane keyboards” in “Language of composite etype percepts” is $Keyboard \sqcap \exists typeSwitch.(Mechanical \sqcup Membrane)$
5. The formalization of “The set of all mechanical and membrane keyboards” in “Language of composite etype percepts” is $Keyboard \sqcap \exists typeSwitch.(Mechanical \sqcap Membrane)$

Exercise 1.9 Indicate which of the formalizations in the different sublanguages used by the Logic of Descriptions (LOD) of the following sentences in natural language are correct (one or more):

1. The formalization of “A doctor is a person who works in a hospital” in “Language of composite etype percepts” is $Doctor \sqsubseteq Person \sqcap \exists worksIn.Hospital$
2. The formalization of “Smartphones are exactly those devices that can access the internet” in “Language of composite etype percepts” is $Smartphone \equiv Device \sqcap \forall canAccess.Internet$
3. The formalization of “Entities that have neither a cat nor a dog” in “Language of composite etype percepts” is $\neg(\exists owns.Cat \sqcup \exists owns.Dog)$
4. The formalization of “A backpack is a bag used to carry books” in “Language of composite etype percepts” is $Backpack \equiv Bag \sqcap \exists usedToCarry.Book$
5. The formalization of “A kitchen appliance is an object used for cooking” in “Language of composite etype percepts” is $KitchenAppliance \sqsubseteq Object \sqcap \exists usedFor.Cooking$
6. The formalization of “Runners are the people who practice running” in “Language of composite etype percepts” is $Runner \sqsubseteq Person \sqcap \exists practices.Running$

Exercise 1.10 Indicate which of the formalizations in the different sublanguages used by the Logic of Descriptions (LOD) of the following sentences in natural language are correct (one or more):

1. The formalization of “Entities that study math or physics” in “Language of composite etype percepts” is $\exists studies.Math \sqcap \exists studies.Physics$
2. The formalization of “Entities that study math or physics” in “Language of composite etype percepts” is $\exists studies.Math \sqcup \exists studies.Physics$
3. The formalization of “Vegan dishes are dishes that do not contain any animal product” in “Language of composite etype percepts” is $VeganDish \equiv Dish \sqcap \neg \forall contains.AnimalProduct$
4. The formalization of “Professors are people who teach only at universities” in “Language of composite etype percepts” is $Professor \sqsubseteq Person \sqcap \forall teachesAt.University$
5. The formalization of “A bird is an animal that can fly” in “Language of composite etype percepts” is $Bird \sqsubseteq Animal \sqcap \exists can.Fly$
6. The formalization of “Entities that don’t have a phone” in “Language of composite etype percepts” is $\neg \exists has.Phone$

1.3 Solutions

Solution 1.1

1. $\{Car\#1, Motorbike\#1\}$
2. $\{Bobby\#1, Marly\#1, Toby\#1, Gigi\#1\}$
3. $\{Alice\#1, Anna\#1, Giulio\#1\}$
4. $\{Alice\#1, Giulio\#1, Car\#1, Motorbike\#1, Bicycle\#1, Toby\#1, Bobby\#1, Gigi\#1, Marly\#1\}$
5. $\{Alice\#1, Anna\#1, Giulio\#1, Mario\#1\}$
6. $\{Car\#1, Motorbike\#1, Bicycle\#1, Toby\#1, Bobby\#1, Gigi\#1, Marly\#1\}$
7. $\{\}$

Solution 1.2

1. $\{Mario\#1, Car\#1, Motorbike\#1\}$
2. $\{Mario\#1\}$
3. $\{Toby\#1, Gigi\#1\}$
4. $\{Anna\#1\}$
5. $\{Anna\#1, Mario\#1\}$
6. $\{Mario\#1, Anna\#1\}$
7. $\{\}$

Solution 1.3

1. True
2. True
3. False
4. True
5. False
6. True
7. True

Solution 1.4

1. True
2. True
3. True
4. False
5. False

Solution 1.5

1. True, it is a primitive etype percept and therefore also a composite etype percept
2. True, it is a composition object properties
3. False, this sentence cannot be expressed in the language of composite etype percepts
4. False
5. False, in this sentence “e” (“and”) must be translated as union
6. True, see answer 5
7. True

Solution 1.6

1. False
2. False: The language of composite type perceptions does not contain the symbol of subsumption.
3. True
4. True
5. False

Solution 1.7

1. True, it is a primitive etype percept and therefore also a composite etype percept
2. True, it is a composition object properties
3. False, this sentence cannot be expressed in the language of composite etype percepts
4. False
5. False, in this sentence “e” (“and”) must be translated as union
6. True, see answer 5
7. True

Solution 1.8

1. False
2. False : The language of composite type perceptions does not contain the symbol of subsumption.
3. True
4. True
5. False

Solution 1.9

1. True
2. False
3. True
4. False
5. True
6. True

Solution 1.10

1. False, in this sentence “or” must be translated as union
2. True
3. False
4. True
5. True
6. True